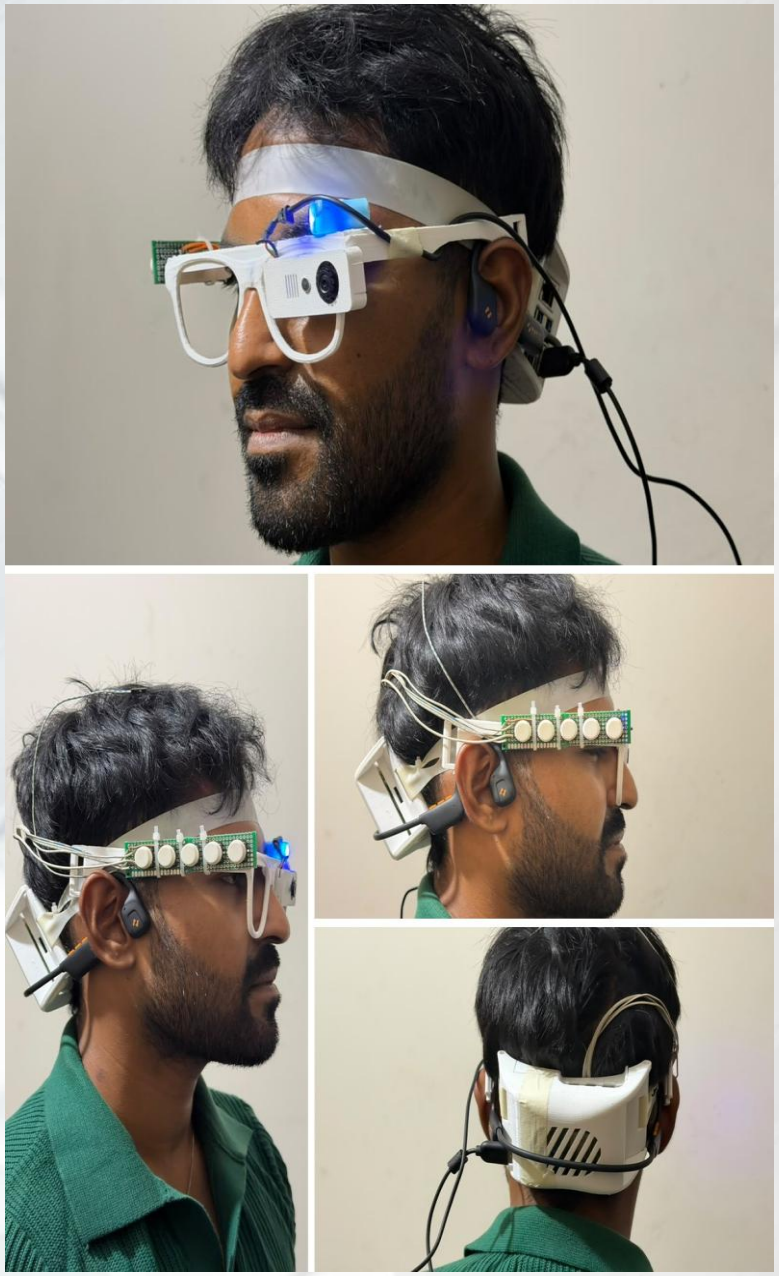
SARA · Smart Assistive Reading Aid

# A Wearable Assistive Glass for Authenticating Bangladeshi Banknotes: Fusing Physical Security Features and Deep Learning for Visually Impaired Users

*A bilingual (Bangla and English) smart glass that also reads printed Bangla and English text and detects everyday objects, offline on a Raspberry Pi 5*

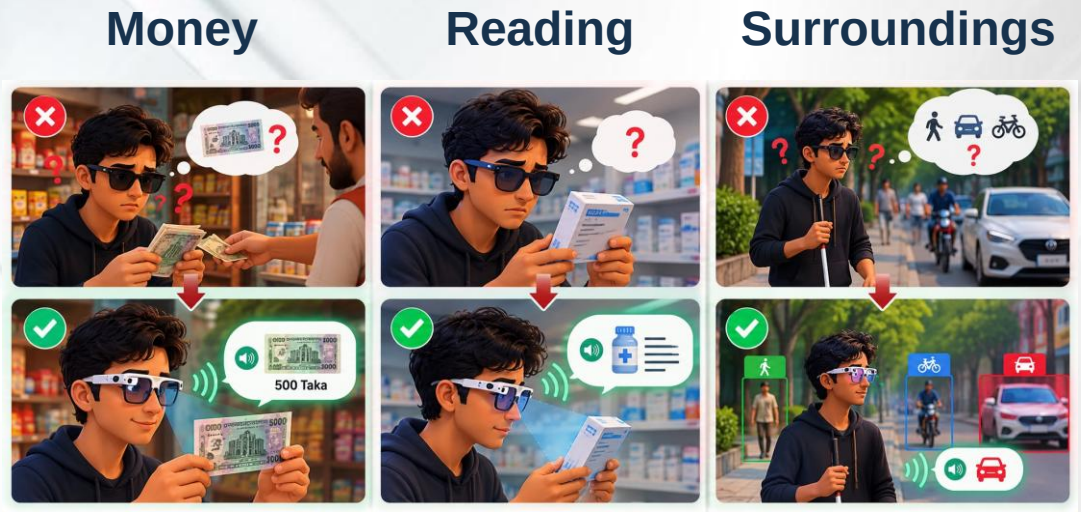

**Supervisor:**
Mohammad Shidujaman, PhD
Assistant Professor
Department of Computer Science & Engineering

**Submitted By:**
MD Abdun Fattah Lam (2220014)
Mukut Protim Memo (2020086)
Tanjila Hossain Tapti (2130561)

# Introduction

- At least 2.2 billion people have a near or distance vision impairment, and its prevalence in low- and middle-income regions is about four times that in high-income regions [1]
- In Bangladesh, three everyday questions usually require help from another person:
  - **Money:** which note was handed back, and is it genuine?
  - **Reading:** what does a medicine label, sign or letter say?
  - **Surroundings:** what or who is ahead?
- In each case the blind person depends on an answer they cannot verify
- **SARA is an offline, bilingual assistive glass built for this setting**




*Top row: today, an answer from another person. Bottom row: with SARA, the glass answers. (Illustration)*

# Problem Statement




## An answer they cannot verify

- A wrong note or a fake note can go unnoticed
- A label or letter can be misread
- Every task waits for someone else

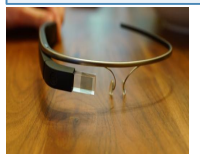
## Existing aids do not close this gap

**OrCam MyEye [2]**



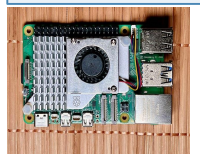
Offline, but no Bangla and no Taka

**Research glasses [3, 4]**



Pictures go to a server

**Pi reading aids [5]**



Mostly English-only

**Counterfeit studies [6, 7, 8, 9, 10]**



Not on a wearable

**The gap**

No reviewed device is offline, speaks Bangla and names Taka notes

# Proposed Solution: SARA

**User wearing SARA (staged)**



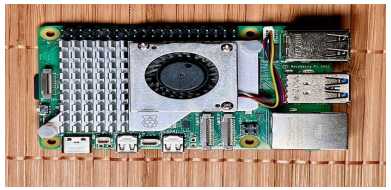
*button, or "Glass, check money" [11]*

**Glass camera**



*captures a picture*

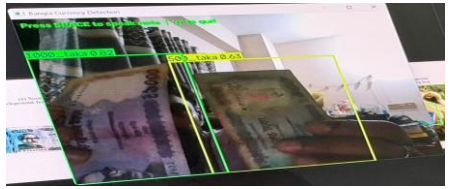
**Raspberry Pi 5 [12]**



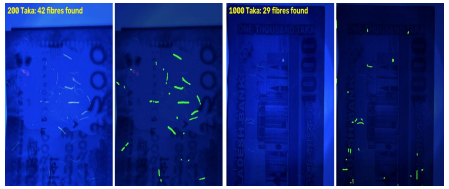
**Offline:** text, objects, Taka, emotion
**Online (optional):** scene description only

*processes offline*

**Taka recognition [13]**



**Authenticity: watermark + UV**



**Text reading [14, 15]**



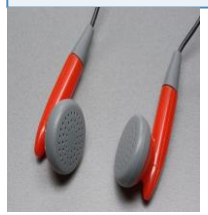
**Objects and people [13, 16]**



**Safe verdict (money)**

"Likely genuine" or "check by hand whether it is genuine"; it never says "counterfeit"

**Spoken answer in Bangla or English [17] + a short vibration**

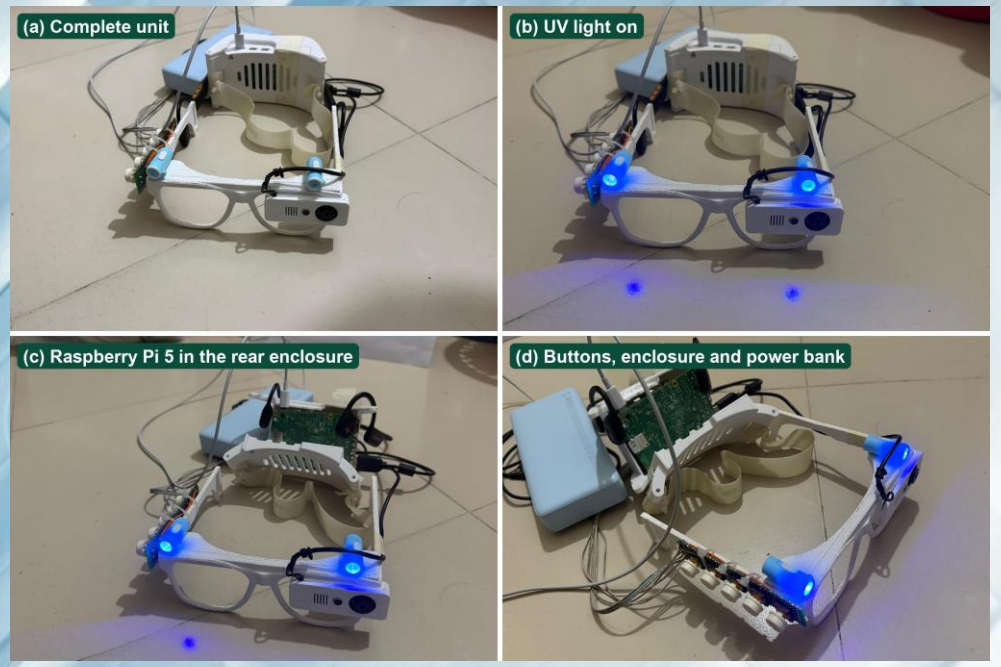





**পাঁচশত টাকার নোট**

"500 Taka note": known without asking anyone

# System Features




*Prototype version 2: the complete unit, the UV light on, the Pi 5 in the rear enclosure, buttons and power bank*

**Text reading**

Bangla and English

**Objects**

Named in Bangla

**Taka notes**

Value + safe check

**Emotion**

Face in front

**Online description**

Optional; needs internet

**How the user starts a mode:** five buttons on the frame, or a voice command: "Glass, read text"

*Models: text [14, 15] CRAFT + EasyOCR; objects [13, 16] YOLOv8 on COCO; emotion [18] RAF-DB; voice commands [11] Vosk.*

# Research Questions

### 1 System

Can it all run on a Raspberry Pi 5?

### 2 Synthetic data

Can unlabelled training data detect real notes?

### 3 Number of views

Can one view of a note be enough?

### 4 Unseen prints

Does the check hold on new counterfeit prints?

### 5 Safety

Can it never accuse a genuine note?

### 6 Bangla OCR

What matters most for reading Bangla?

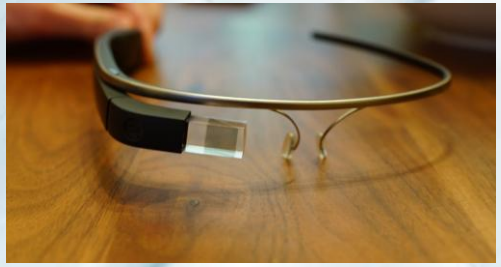
# Literature Review: Existing Assistive Devices

| System | Reads Bangla | Knows Taka | Works offline | Wearable |
|---|---|---|---|---|
| OrCam MyEye [2] | No | No | Yes | Yes |
| Google Glass scene analysis [3] | No | No | No | Yes |
| AIris (2024) [4] | No | No | No | Yes |
| Raspberry Pi reading aids * | No | No | Partly | Partly |
| **SARA (ours)** | **Yes** | **Yes** | **Yes** | **Yes** |

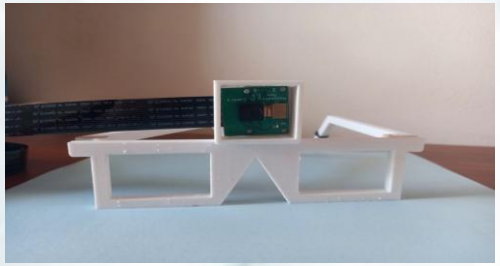
*Rows with a reference: from the cited source. * Our assessment of typical hobby builds, not a cited result.*



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

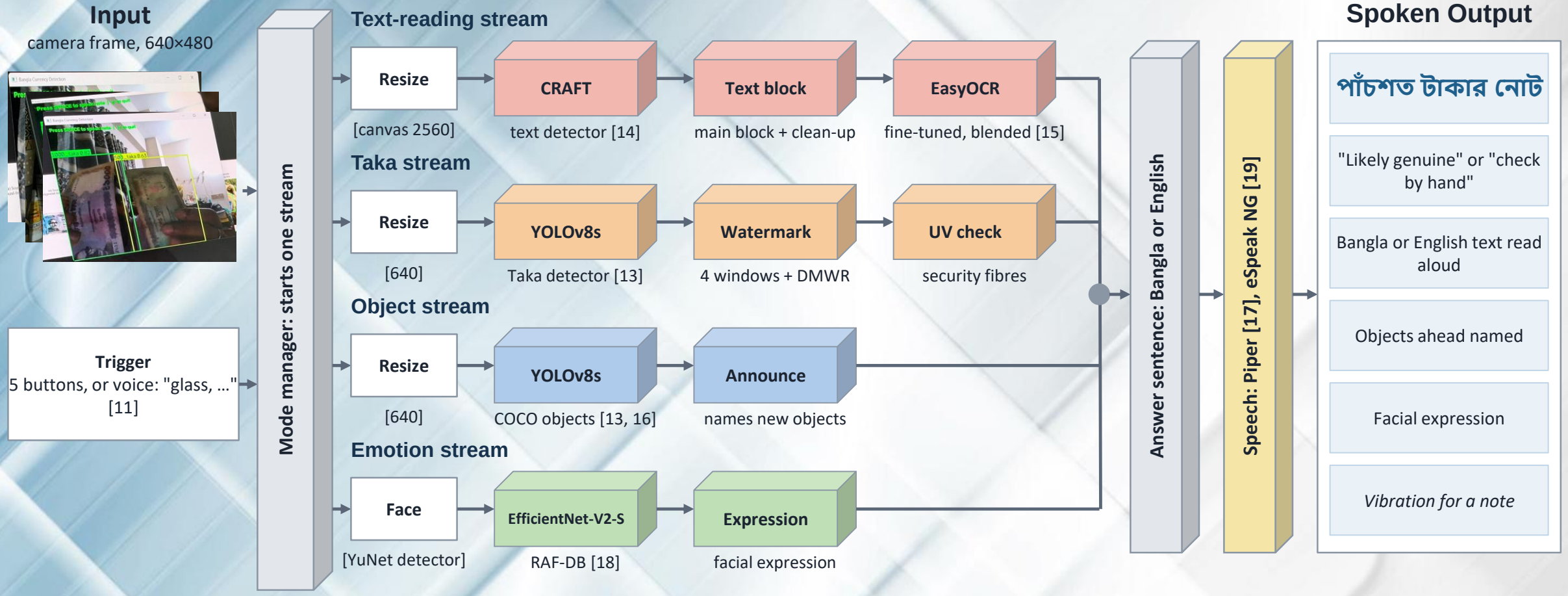
## The research gap

**None of the systems we reviewed combines Bangla, Taka and offline use.**

- Counterfeit tests reuse the same prints
- Few report speed on the real device

*Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].*

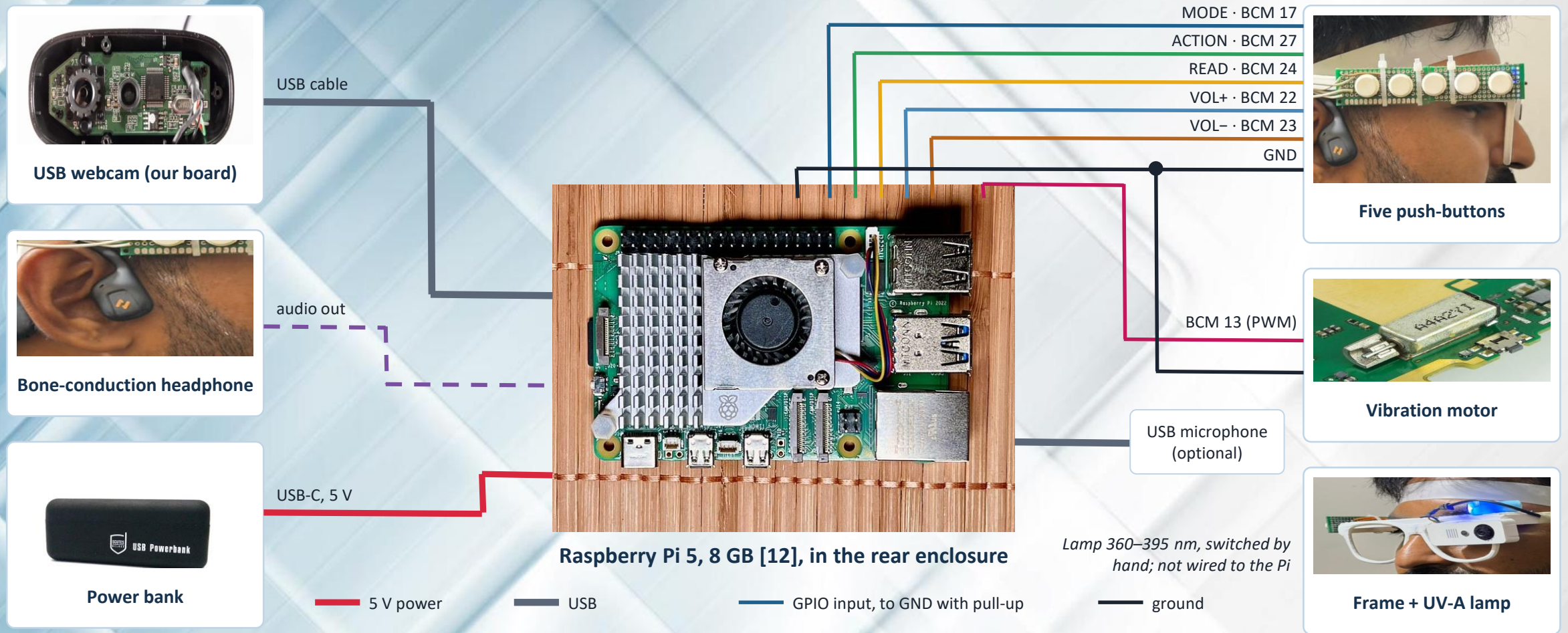
# System Architecture




*All streams run on the Raspberry Pi 5 CPU, offline. The optional online scene-description mode is not shown.*

*Sources: [11] Vosk; [13] Ultralytics YOLOv8; [14] Baek et al. 2019; [15] EasyOCR; [16] COCO; [17] Piper; [18] RAF-DB; [19] eSpeak NG.*

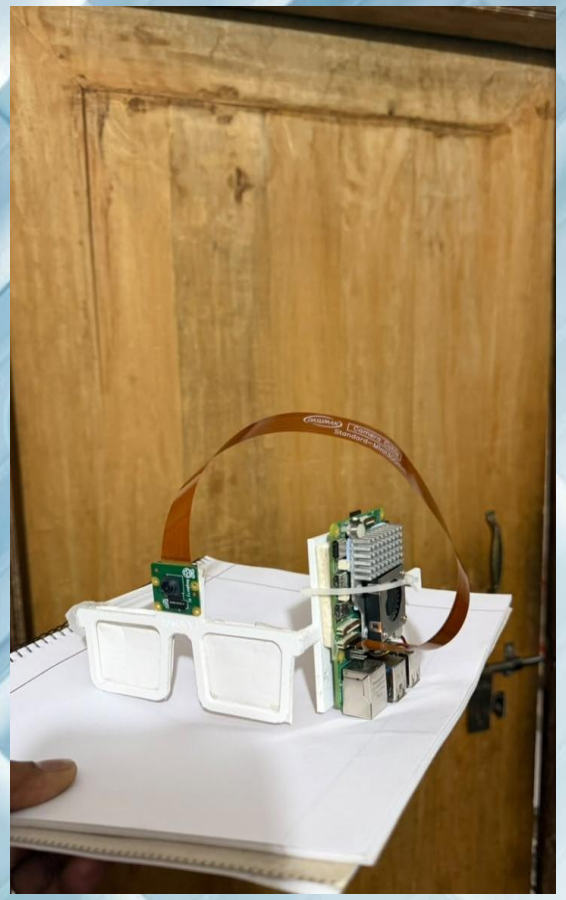
# Hardware Design




*Schematic, not to scale; pin numbers are those set in the code.*

*Pictures (Wikimedia Commons): Raspberry Pi 5 by SimonWaldherr, CC BY 4.0; vibration motor by Raimond Spekking, CC BY-SA 4.0; power bank by Pinguinguy, CC0. Webcam board, buttons, headphone and frame: our prototype.*

# Evolution of the Prototype



**Version 1**
Board on one temple




**Front**
Camera + UV lamp



**Side**
5 buttons, bone-conduction earphone



**Rear**
Pi 5 in a ventilated case

# Banknote Detection: Synthetic Training Data




Note photos [20] are pasted onto real scenes [16], and the program writes every label itself.

**SYNTHETIC COMPOSITES**

**0.995**

mAP@0.5 on 2,282 held-out composites

**REAL PHOTOS**

**94.1%**

right note value on 1,536 photos taken by other people (laptop)

*Sources: [20] BanglaTaka (Mendeley); [16] Lin et al., COCO, ECCV 2014; detector [13] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).*

# Banknote Detection: Hand-Held Notes

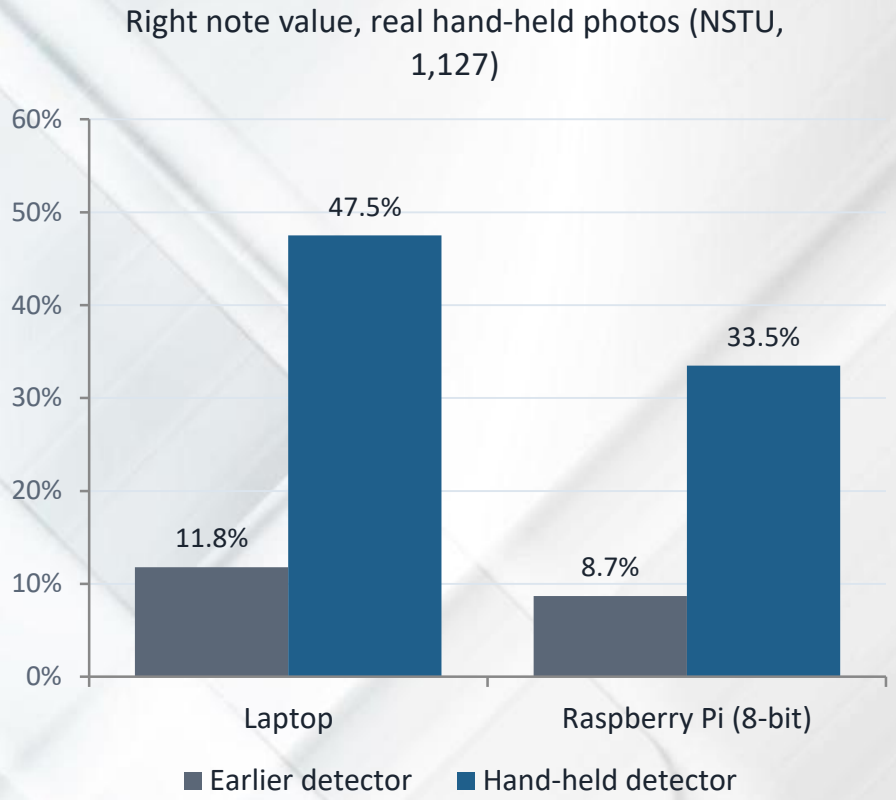
**TRAINING: 6,000 SYNTHETIC CLOSE-UPS**



**TEST: REAL HAND-HELD PHOTOS**

# 11.8% → 47.5%

*Since 29 Oct the Pi runs the laptop's full-precision detector. Trade-off: with 3 notes in view, the total is right 97.0% (was 99.2%).*

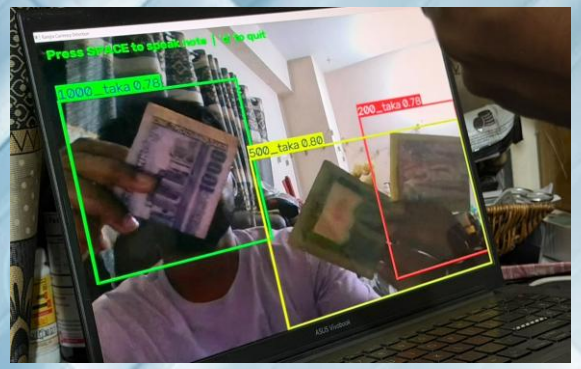


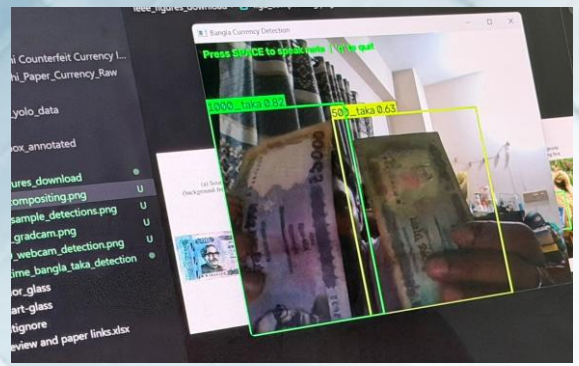

*Sources: test photos [21] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [13] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.*

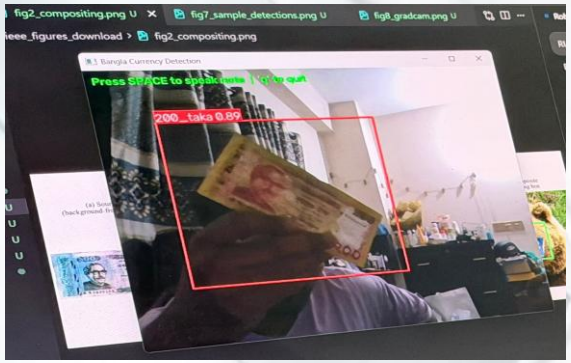
# Live Detection Tests

**Informal demonstrations, not a measured accuracy study**

## Software: live detection on a laptop webcam (detector output shown)







## Second prototype (v2): live tests with the glass (staged)



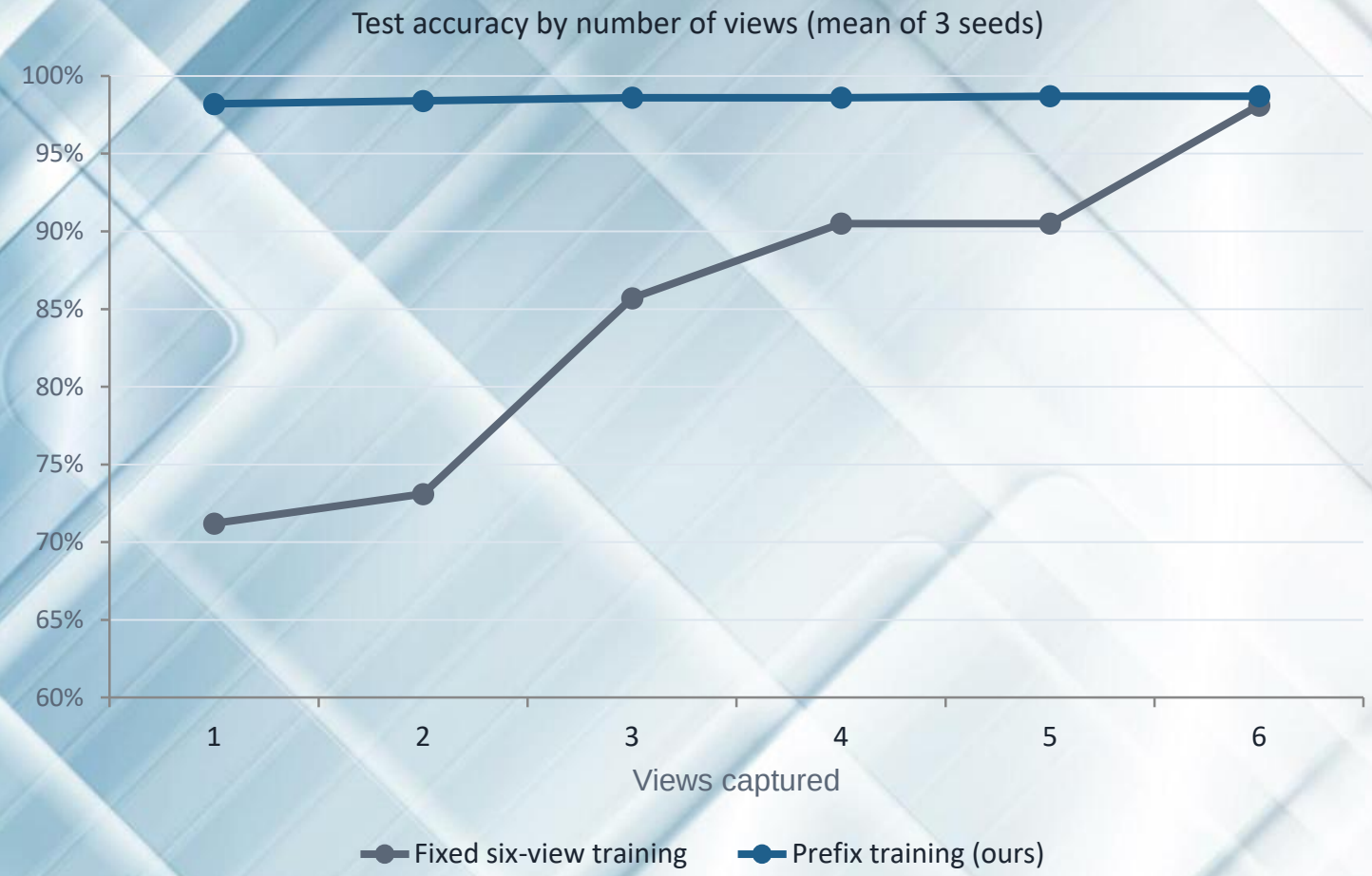






Screenshots: 50, 200, 500 and 1,000 Taka named at 0.63–0.89 confidence, two or three notes at once. The v2 photos do not show the detector output.

# Counterfeit Screening: Prefix-Robust Multi-View Training




With a single view

**71.2%**

*rises to*

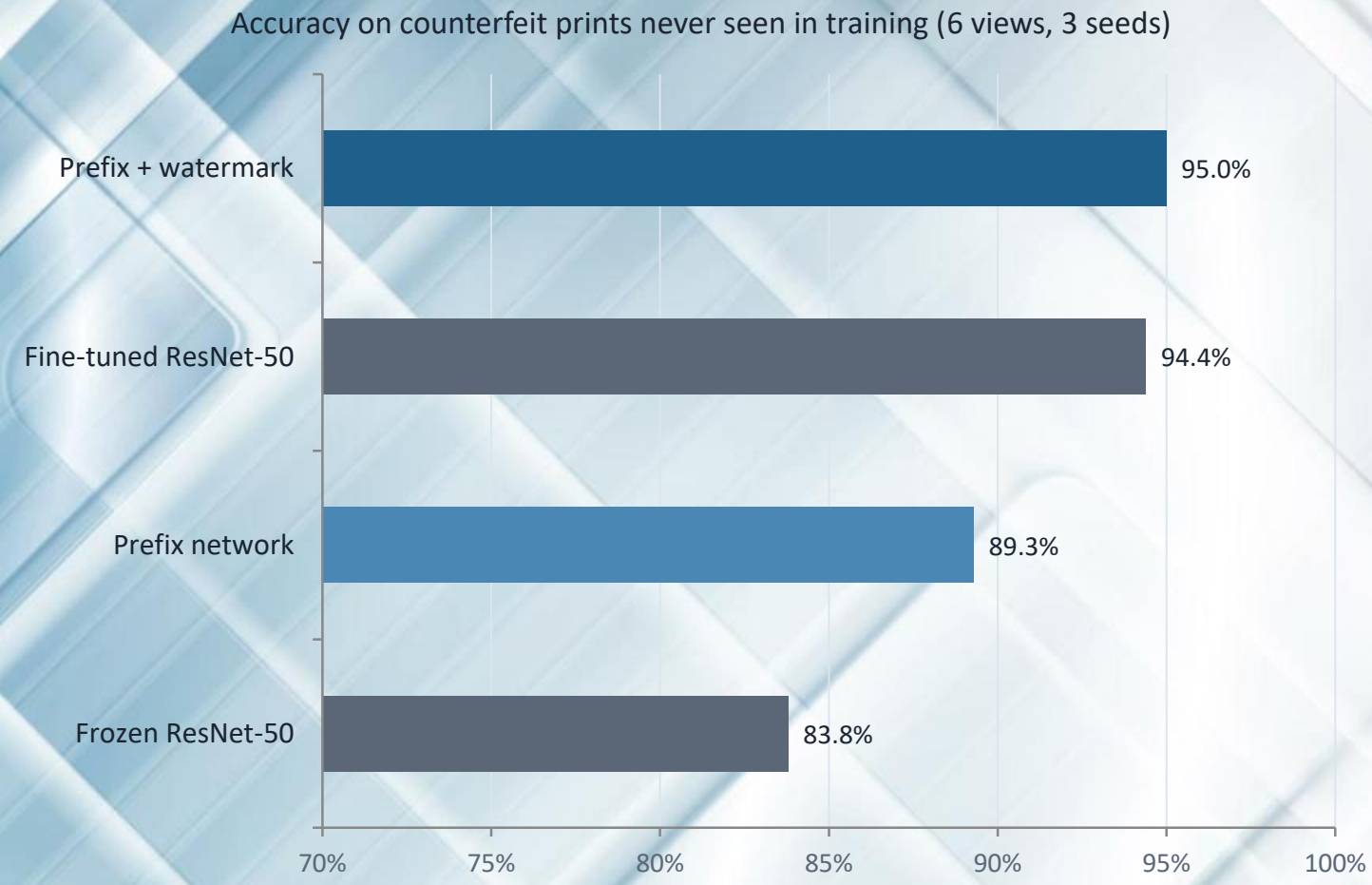
**98.2%**

Training on random first views: one view is enough.

*Held-out JaalTaka test notes, never used for training or for any choice*

*Data: [9, 10] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).*

# Counterfeit Screening: Evaluation on Unseen Prints



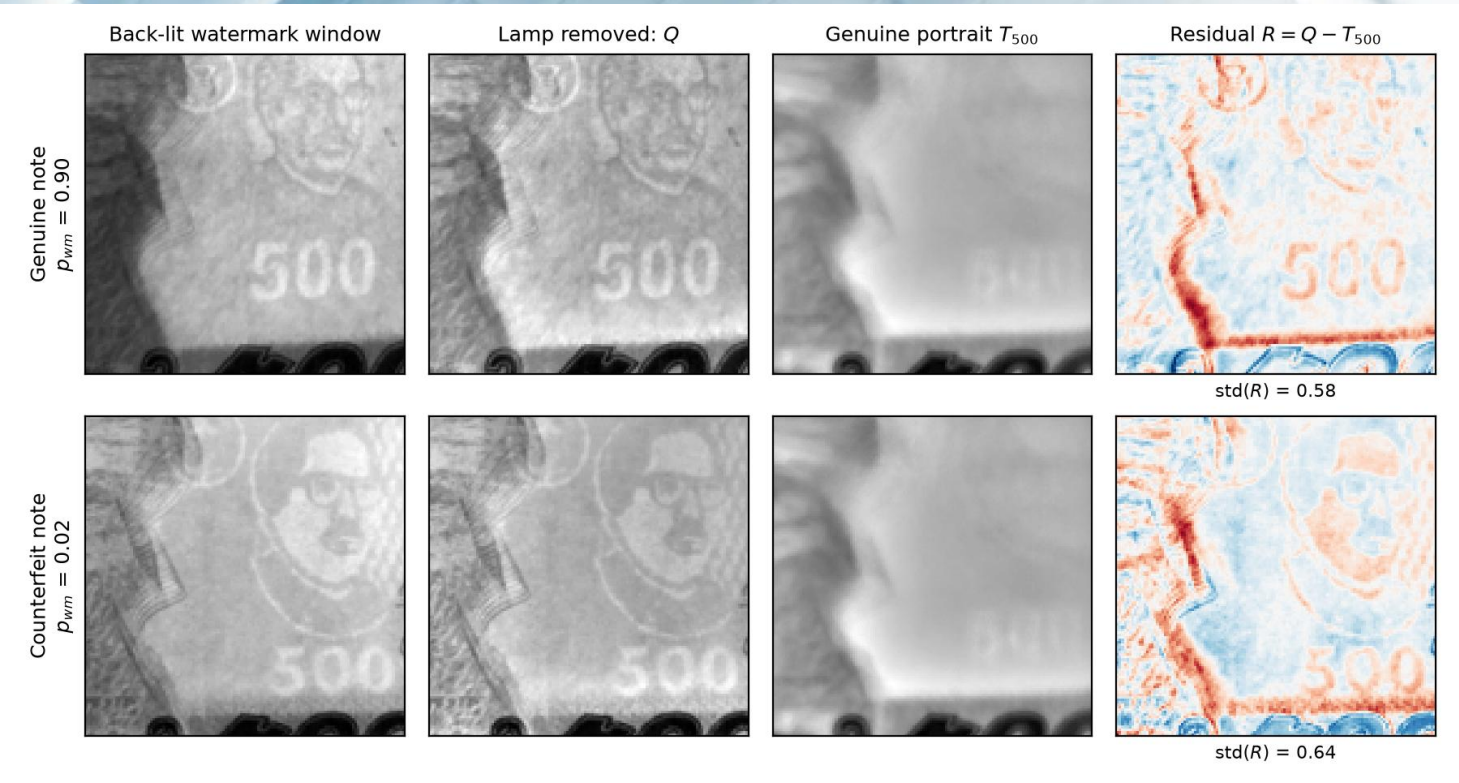

**95.0%**

on counterfeit prints never seen in training, with the watermark check

**Why a stricter split:** 279 of 322 readable 500 Taka counterfeits share one serial number, so the usual split tests copies of known prints.

*Data: [9] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [6] SVM on security features; [7] CNNs; [8] UV flecks + RPN.*

# Watermark Verification: DMWR




1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

**91.5%**

on unseen counterfeit prints (3 seeds, ±0.3)

**46× smaller**

than MobileNetV2 [22] (47,818 parameters, 0.21 MB), and as accurate: 92.9%, p ≥ 0.375

*Baseline [22] Sandler et al., MobileNetV2, CVPR 2018; data [9] JaalTaka. DMWR (denomination-matched watermark residual) is our method.*

# Safe Verdict Policy




**0 / 88**

test counterfeits passed as "likely genuine"

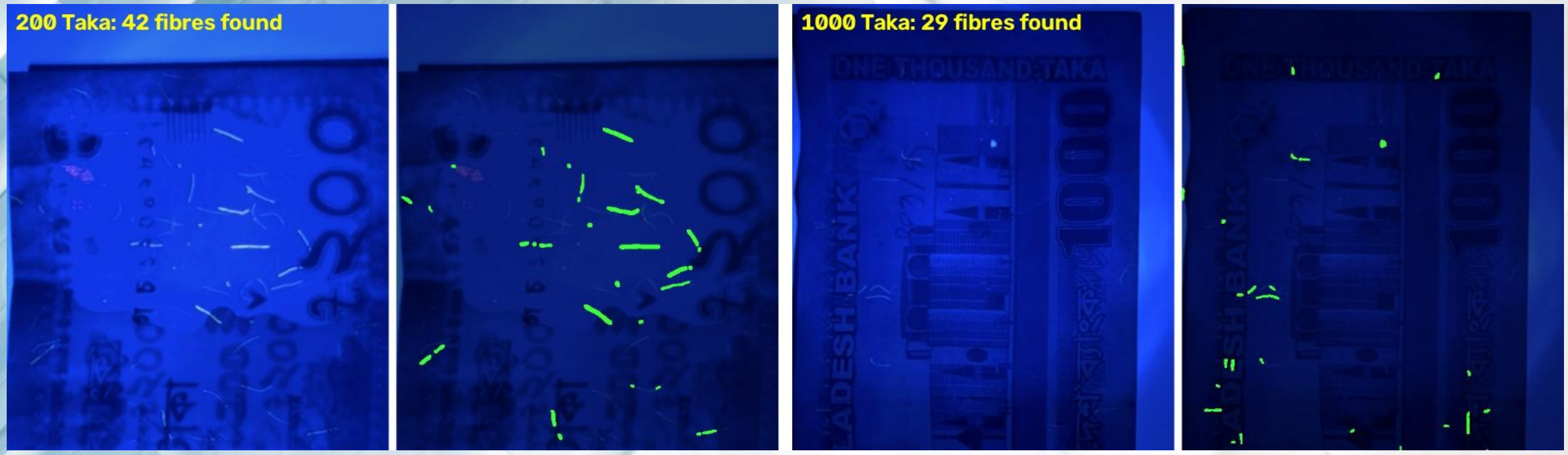
**"Likely genuine":** the score is higher than every counterfeit seen in validation (57.5% of genuine notes).

**"Check by hand":** the glass is not sure. It never means "counterfeit".

**A screening aid, not a guaranteed counterfeit detector**

*Test notes: [9] JaalTaka.*

# UV Security-Fibre Verification




**85.6%**

of 111 real UV photos of genuine notes accepted

**88.6%**

of SIMULATED fibre-free sheets rejected (not real counterfeits)

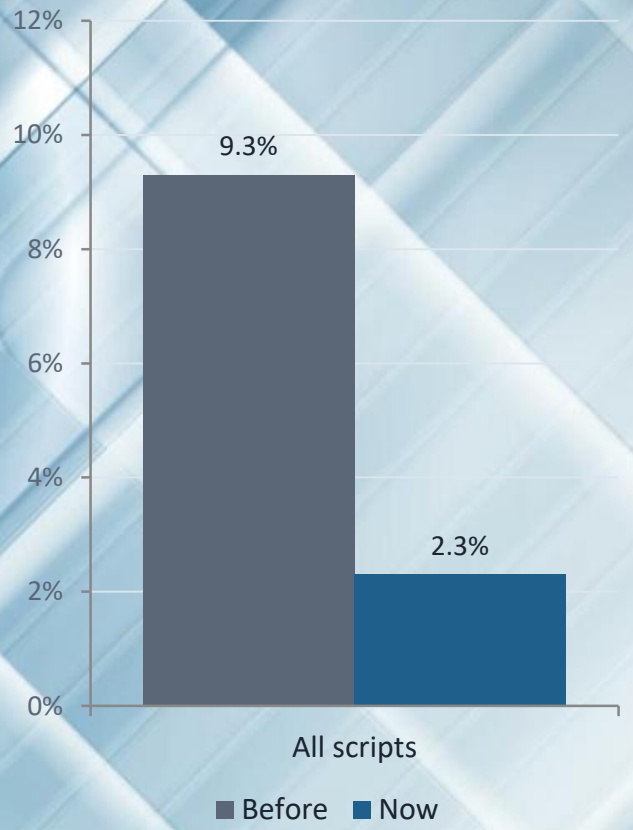
**Speaks under the lamp**

"Likely genuine" or "check by hand", never "counterfeit". Not yet tested on real counterfeits or through the glass camera

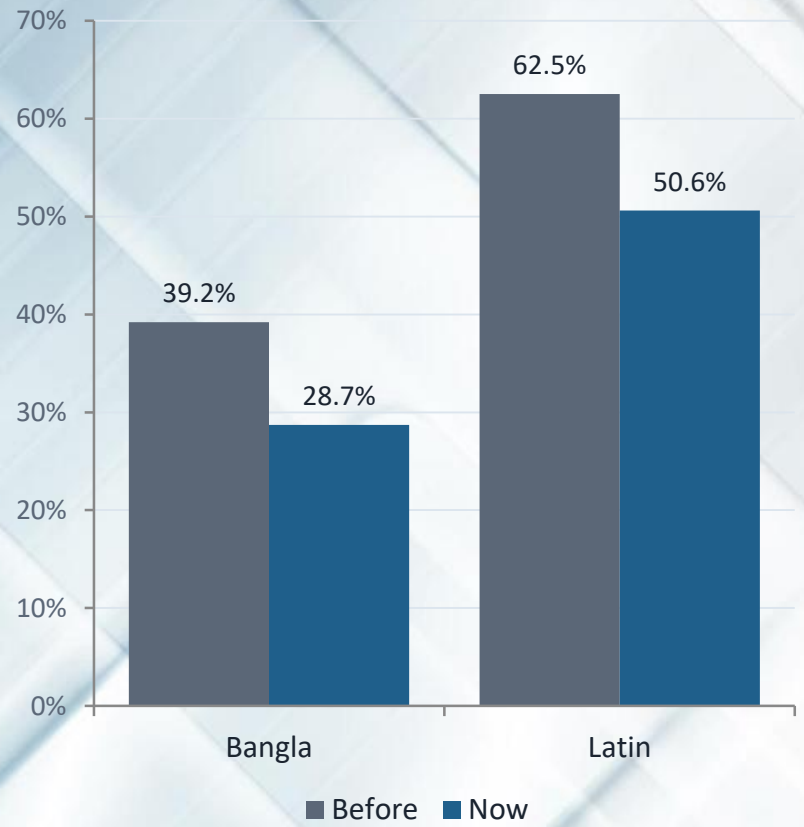
*The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [23]. Prior Taka UV work: [8] (96.3%). Fibre check and photos are ours.*

# Bangla and English Text Recognition

## Rendered text (clean)




## Real word photos (hard)




## What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words
- Sharpest frame of 1 s: shaken text 28.0% → 3.8%

## What did not

Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

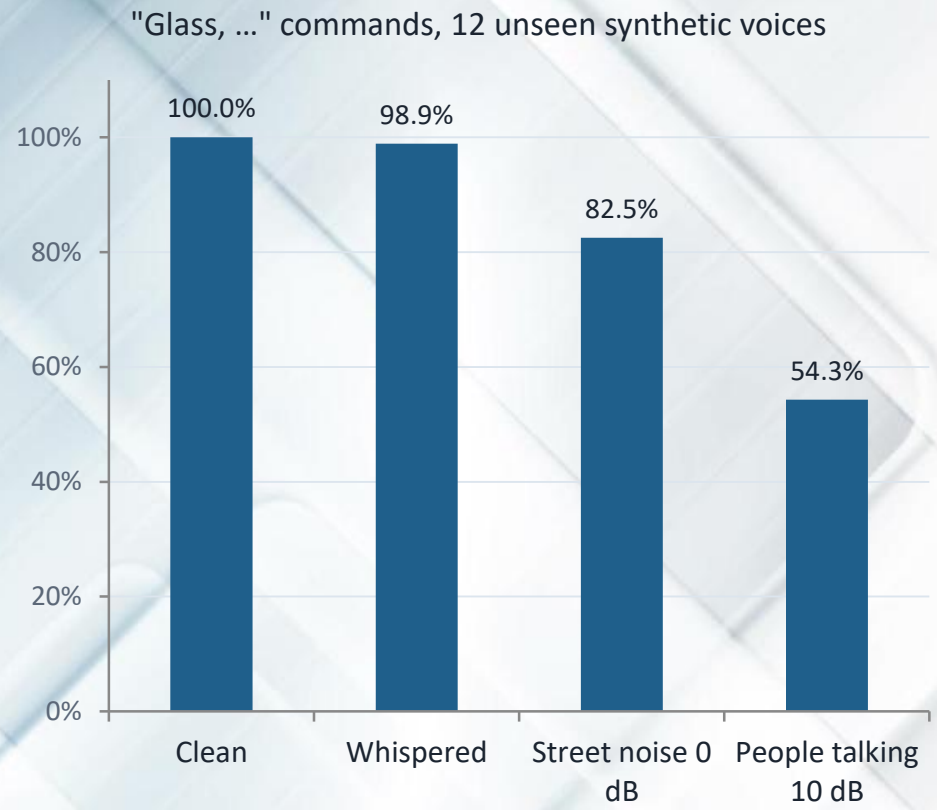
***Character error rate, lower is better. Measured on evaluation datasets, not yet through the glass.***

*Engine [15] EasyOCR with [14] CRAFT; real words from [24] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.*

# Voice Interaction

## Speaking

- Natural Piper voice, Bangla and English [17, 19]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured




**212 → 13**

false commands in 5,040 ordinary sentences, thanks to the wake word

**0.86 s**

from the end of speech to the action (median)

**Command results: synthetic voices on a laptop, not human trials on the glasses**

*Recogniser [11] Vosk; noise from [25] Piczak, ESC-50; voices [17, 19]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.*

# Performance on the Raspberry Pi 5

**422 ms**
currency check, end to end

**51 ms**
watermark check

**462 ms**
object scan (YOLOv8s)

**536 ms**
emotion, face in view

**5.4 s**
3-line sign (18.7 s on 1 October)

**2.76 GB**
peak memory of the app (of 8 GB)



**Board temperature**

**57.8–62.3 °C**

measured on the board with a thermometer while the glass was running

*The processor's own sensor reads higher: in 30 minutes of non-stop load it peaked at 87.0 °C, with the soft limit active in 55% of samples.*

*Timings from the 29 October 2026 benchmark on the Pi 5 (8 GB). The Pi now runs the full-precision note detector: about 0.3 s more per check, not yet timed.*

*Board: [12] Raspberry Pi 5 (8 GB), fan and heat sink, C270 webcam. All timings are ours (benchmark_report/2026-10-29/); board-surface temperature measured by the team with a thermometer.*

# Prototype Testing and Project Status




*Prototype version 2 in a staged shop exchange by team members; illustration only, no error rate counted.*

**DONE** **Engineering checks**

Each part tested on held-out data, on a laptop and the Pi 5

**PRELIMINARY** **First field test**

Glass camera: currency unreliable, Bangla text not read

**IN PROGRESS** **Participant trials**

Sessions with a visually impaired participant have started

**NONE YET** **Full user study**

No results about users are claimed

# Future Directions




## Camera

Higher resolution or autofocus

## User study

8–12 participants, including their own voice commands

## More banknotes

Unseen counterfeit prints; UV check on real fakes

## Speed and cooling

Time the new code on the Pi; better cooling; battery

**Done:** engineering tests **In progress:** participant trials **Future:** the four steps above

# Conclusion



## Built

An offline Raspberry Pi 5 glass that reads Bangla and English, names objects and Taka notes, and speaks Bangla

## Investigated

Synthetic training data, prefix training, unseen prints, a safe verdict, DMWR, hand-held notes, OCR, voice

## Still to validate

Real counterfeits (also under UV), people's voices, the glass camera, a completed user study

# THANK YOU

Questions?

# References


[1] World Health Organization. Blindness and vision impairment: fact sheet, updated 10 February 2026.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Adusumilli, V. et al. Reading aid and translator with Raspberry Pi for blind people. ICACCS, 2023, pp. 327-331.

[6] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[7] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[8] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[9] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[10] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[11] Alpha Cephei. Vosk offline speech recognition toolkit.

[12] Raspberry Pi Ltd. Raspberry Pi 5 product brief, 2023.

[13] Jocher, G., Qiu, J., Chaurasia, A. Ultralytics YOLO, version 8.0.0, 2023.

[14] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[15] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[16] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[17] Rhasspy. Piper: a fast, local neural text-to-speech system.

[18] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[19] eSpeak NG text-to-speech.

[20] Nuhash, M. N. I., Akter, S., Mojumdar, M. U. A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data V1, 2024.

[21] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[22] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[23] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[24] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[25] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

# Resources and Links

## Our code and results

**SARA: glass software for the Raspberry Pi 5**
github.com/aflam330/SARA

**Research repository: experiments, results, thesis**
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

## Datasets

**JaalTaka (counterfeit notes)**
data.mendeley.com/datasets/2m7wk5cy4c/2

**BanglaTaka (note photos)**
data.mendeley.com/datasets/3cv2sypkkh/1

**NSTU-BDTAKA (hand-held notes)**
www.sciencedirect.com/science/article/pii/S2352340924006681

**Microsoft COCO**
cocodataset.org

**ICDAR 2019 MLT**
rrc.cvc.uab.es/?ch=15

**ESC-50 environmental sounds**
github.com/karolpiczak/ESC-50

## Open-source tools

**Ultralytics YOLOv8**
github.com/ultralytics/ultralytics

**EasyOCR**
github.com/JaidedAI/EasyOCR

**Piper text-to-speech**
github.com/rhasspy/piper

**Vosk speech recognition**
alphacephei.com/vosk/

**eSpeak NG**
github.com/espeak-ng/espeak-ng

**WHO vision fact sheet**
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment